Classical Management and the Future of US Finance: Taylor, Fayol, and Agentic AI in US Financial Institutions

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Abstract

As American financial institutions race to maintain global competitiveness in the age of artificial intelligence, the classical management frameworks of Frederick Winslow Taylor and Henri Fayol offer unexpected yet vital guidance for navigating this transformation. This paper examines the complementary relationship between Taylor's scientific management—focused on task-level efficiency through systematic measurement and division of labor—and Fayol's administrative theory—emphasizing enterprise-wide coordination through planning, organizing, and controlling functions. Together, these foundational frameworks provide a comprehensive lens for understanding how US banks, hedge funds, and financial institutions can effectively integrate agentic AI systems into their operations. The paper introduces a novel framework conceptualizing generative AI agents as embodying Taylor and Fayol functions: some agents optimize specific tasks (Taylor agents) while others coordinate across agents and functions (Fayol agents). This multi-agent architecture is transforming American banking through automated trading, risk management, customer service, and compliance—areas

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critical to US financial competitiveness. The analysis draws on recent scholarship examining AI's role in teamwork, resilience, decision-making (Joshi, 2025e), leadership and management trends (Joshi, 2025b), the convergence of AI and emotional intelligence (Joshi, 2025a), and quantitative models for managerial decision-making (Joshi, 2025c). The paper reveals that classical management theories remain relevant but require reconceptualization when intelligent agents, rather than humans, perform both specialized and coordinative functions. The paper concludes with implications for American managers and financial institutions adopting agentic AI systems to secure the nation's competitive edge in global financial markets.

1 Introduction

Management represents a vital function in all organizations, from manufacturing floors to quantitative hedge funds. The foundational works of Henri Fayol (1841–1925) and Frederick Winslow Taylor (1856–1915) established principles that continue to shape management theory and practice more than a century later. Fayol, a French mining engineer and director, defined the main functions of management as planning, organizing, coordinating, and controlling (Fayol, 1949). Taylor, an American engineer and consultant, developed scientific management principles that applied engineering techniques to work processes to maximize efficiency (Taylor, 1911). As recently as 2001, Taylor's work was voted the most influential management book of the twentieth century, demonstrating the enduring relevance of these classical frameworks.

The contemporary business environment, however, differs dramatically from the industrial context in which Taylor and Fayol developed their theories. The rise of quantitative finance and artificial intelligence has transformed how organizations operate, make decisions, and manage workers. Quantitative hedge funds such as Renaissance Technologies and Two Sigma employ sophisticated mathematical models and machine learning algorithms to execute trades, often with minimal human intervention (Zuckerman, 2019). AI systems now perform tasks that previously required human judgment, including risk assessment, portfolio optimization, and even some aspects of management itself.

Most significantly, the emergence of **agentic generative AI**—autonomous AI agents capable of perceiving their environment, making decisions, and taking actions to achieve specific goals—is fundamentally reshaping banking and finance. These agents are not passive tools but active participants in organizational processes. They can be designed to embody different management functions: some agents specialize in optimizing specific tasks (functioning as digital ”Taylors”), while others coordinate across multiple agents and functions (functioning as digital ”Fayols”). This multi-agent architecture creates a new organizational form where management functions are distributed across a network of intelligent agents that communicate, negotiate, and collaborate.

Recent research has extensively reviewed these developments across multiple dimensions. Joshi (Joshi, 2025d) provides a comprehensive review of artificial intelligence in management, leadership, decision-making, and collaboration, offering an organizational behavior-focused synthesis that bridges classical management theory with contemporary AI applications. This work examines how AI systems are reshaping fundamental management processes and identifies the capabilities managers need to develop in AI-augmented organizations. Similarly, Joshi (Joshi, 2025c) reviews quantitative models and visualization approaches for managerial decision-making in the age of AI, demonstrating how data-driven methods can support and enhance managerial judgment.

The role of AI in enhancing teamwork, resilience, and decision-making has been systematically analyzed (Joshi, 2025e), revealing that hybrid AI-human decision models achieve significant improvements in response times while maintaining high prediction accuracy in behavioral assessments. Furthermore, the convergence of artificial intelligence and emotional intelligence has profound implications for leadership and organizational behavior (Joshi, 2025a), particularly in areas such as emotion recognition systems, AI-powered feedback tools, and the development of hybrid competencies that combine technical AI fluency with advanced emotional intelligence skills. Joshi (Joshi, 2025b) identifies three key areas of AI impact on leadership: enhanced strategic decision-making through human-AI collaboration, evolution of leadership styles in digital environments, and organizational challenges in AI adoption.

This paper addresses a critical question: How do the classical management frameworks of

Taylor and Fayol apply to organizations where agentic generative AI systems perform both specialized and coordinative functions? The analysis proceeds in four parts. First, it examines the complementary relationship between Taylor's scientific management and Fayol's administrative theory. Second, it reviews the required readings on management theory and practice, integrating contemporary AI scholarship. Third, it introduces a framework for understanding agentic AI in banking through the Taylor-Fayol lens, with specific applications to trading, risk management, and customer service. Fourth, it discusses implications for managers and suggests directions for future research.

2 Visual Frameworks: Mapping Classical Theory to Agentic AI

To better illustrate the conceptual relationships between classical management theories and modern agentic AI systems, this section presents a series of visual frameworks through detailed textual descriptions. These frameworks synthesize the key arguments of the paper and provide a conceptual representation of the Taylor-Fayol agent architecture in quantitative finance.

2.1 Conceptual Overview: From Classical Foundations to Agentic AI

This framework illustrates the paper's core argument: classical management theories evolve into specialized AI agents. Taylor's focus on task-level optimization manifests as Taylor Agents that perform specialized functions, while Fayol's emphasis on coordination manifests as Fayol Agents that integrate and direct these specialized agents. Together, they form multi-agent systems that power key banking functions.

Conceptual Framework: From Taylor and Fayol to Agentic AI

Classical Management Foundations →
Taylor's Scientific Management
Fayol's Administrative Theory

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Task-Level Optimization
(Measurement & Efficiency) Enterprise-Level Coordination
(Planning & Control)

↙ ↘

Taylor Agents
Specialized AI Task Optimizers **Fayol Agents**
AI Coordinators

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Multi-Agent AI Systems in Quantitative Finance

Algorithmic Trading Risk Management Customer Service Compliance Monitoring

Figure 1: Conceptual Framework: From Taylor and Fayol to Agentic AI in Banking (Text Representation)

2.2 Taylor Agents vs. Fayol Agents: A Functional Comparison

Figure 2 provides a detailed comparison of Taylor Agents and Fayol Agents, illustrating their distinct yet complementary roles in multi-agent systems.

Taylor Agents focus on specialized task optimization through measurement and feedback, while Fayol Agents ensure system-wide coordination and alignment with organizational objectives. The information flow between agents creates a structured hierarchy that maintains order while enabling efficient collaboration.

2.3 Agent Communication and Fayol's Scalar Chain

Figure 3 illustrates how Fayol's principle of the scalar chain manifests in multi-agent AI systems through hierarchical communication patterns.

The diagram shows hierarchical patterns where Taylor Agents report upward to Fayol Agents, which issue directives downward. Lateral "gangplank" communications between peer agents en-

Taylor Agents and Fayol Agents: Functional Architecture

Taylor Agents (Task Optimizers)	**Fayol Agents (Coordinators)**
Algorithmic Trading (Execution Optimization)	Portfolio Coordination (Capital Allocation)
Credit Underwriting (Risk Assessment)	Risk Aggregation (Exposure Monitoring)
Fraud Detection (Pattern Recognition)	Compliance Monitoring (Regulatory Adherence)
Customer Service (Query Resolution)	Orchestration Agents (Workflow Management)
Information Flow Taylor Agents → Fayol Agents (Reporting) Fayol Agents → Taylor Agents (Directives) Fayol Agents ↔ Fayol Agents (Coordination)	

Figure 2: Functional Architecture: Taylor Agents and Fayol Agents in Banking AI Systems (Text Representation)

able efficient coordination while maintaining hierarchical integrity—a direct application of Fayol's original insight to agentic architectures.

2.4 AI-Driven Trade Execution: A Multi-Agent Workflow

Figure 4 provides a detailed workflow of the AI-driven trade execution case discussed in Section 4.5, illustrating how multiple agents collaborate in real-time.

This workflow demonstrates the dynamic interactions between Taylor Agents (execution optimization) and Fayol Agents (coordination and control) in a real-world trading scenario, achieving what neither could accomplish alone.

2.5 Human-Agent Collaboration: The Evolving Role of Managers

Figure 5 presents a framework for understanding the changing role of human managers in organizations with agentic AI systems.

As AI agents assume both specialized and coordinative functions, human managers transition

Agent Communication Hierarchy and Fayol's Scalar Chain

Level 1: Strategic Coordination
• Fayol Agent: Portfolio Coordinator
• Fayol Agent: Risk Manager
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Level 2: Tactical Coordination
• Fayol Agent: Compliance Monitor
• Fayol Agent: Orchestrator
↓
Level 3: Task Execution (Taylor Agents)
• Equity Trading Agent • Derivatives Trading Agent
• Credit Scoring Agent • Fraud Detection Agent
• Customer Service Agent

Lateral Communication (Gangplanks):
Taylor Agents may communicate laterally when authorized:
Equity Trading Agent ↔ Derivatives Trading Agent
Credit Scoring Agent ↔ Fraud Detection Agent

Figure 3: Agent Communication Hierarchy and Fayol's Scalar Chain (Text Representation)

from direct supervision to higher-level functions including system design, boundary setting, exception handling, performance evaluation, and emotional intelligence leadership.

2.6 Integration of Emotional Intelligence with AI Systems

Figure 6 illustrates the convergence of artificial intelligence and emotional intelligence as discussed in (Joshi, 2025a).

Hybrid competencies emerge at the intersection of technical AI fluency and advanced emotional intelligence skills, creating new capabilities for human-AI collaboration in areas such as conflict resolution, team motivation, and change management.

2.7 Summary: Integrated Framework

Figure 7 provides an integrated summary of the paper's theoretical contribution, showing how classical management theory combines with contemporary AI scholarship to produce the Taylor-

Multi-Agent Workflow for AI-Driven Trade Execution

Step 1: Initiation

Trading Objective (Purchase Large Block) $\rightarrow$ Trade Coordinator (Fayol Agent)

Step 2: Task Distribution

Trade Coordinator divides objective among Taylor Agents:

- Liquidity Detection Agent
- Execution Timing Agent
- Venue Selection Agent

Step 3: Parallel Analysis

Taylor Agents analyze market conditions and report findings

Step 4: Coordination & Risk Management

- Market Impact Agent aggregates Taylor Agent reports
- Risk Management Agent monitors exposure limits
- Trade Coordinator adjusts execution plan based on feedback

Step 5: Execution & Monitoring

- Taylor Agents execute trades according to adjusted plan
- Transaction Cost Analysis Agent measures performance
- Continuous feedback loop until trade completion

Step 6: Completion

Trade Complete with Optimal Execution

Figure 4: Multi-Agent Workflow for AI-Driven Trade Execution (Text Representation)

Fayol Agent Framework.

These visual frameworks collectively illustrate the paper's core arguments: the complementary relationship between Taylor and Fayol's theories, their manifestation in agentic AI systems, the hierarchical and lateral communication patterns in multi-agent architectures, the evolving role of human managers, and the integration of emotional intelligence with AI capabilities. Together, they provide a comprehensive conceptual summary of how classical management foundations inform and illuminate the design of AI-driven organizations in quantitative finance.

3 The Complementary Works of Taylor and Fayol

Taylor and Fayol developed their management theories independently and from different perspectives, yet their works complement each other in ways that provide a comprehensive foun-

The Evolving Role of Human Managers in Agentic AI Organizations

Traditional Role	Transition Phase	Future Role
Direct Supervision	Hybrid Oversight	**System Designer**
Task Assignment	AI System Training	Architecting Agent Relationships
Performance Monitoring	Exception Management	**Boundary Setter**
Coordination		Defining Autonomy Limits
		Exception Handler
		Managing Edge Cases
		Performance Evaluator
		Measuring System Effectiveness
		Emotional Intelligence Leader
		Fostering Human-AI Collaboration

Figure 5: The Evolving Role of Human Managers in Agentic AI Organizations (Text Representation)

dation for understanding management. Taylor focused on the operational level—how individual tasks could be optimized through scientific analysis—while Fayol addressed the organizational level—how managers should coordinate activities across the enterprise.

3.1 Taylor's Scientific Management

Frederick Winslow Taylor's scientific management emerged from his observations of industrial workplaces in the late nineteenth century. Taylor observed that workers often performed tasks inefficiently, relying on rule-of-thumb methods rather than systematically optimized approaches. His principles of scientific management sought to remedy this through four key elements: (1) developing a science for each element of work to replace old rule-of-thumb methods, (2) scientifically selecting and training workers rather than allowing them to choose their own methods, (3) cooperating with workers to ensure work is done according to scientific principles, and (4) dividing work

The Convergence of AI and Emotional Intelligence in Organizations

Artificial Intelligence	Emotional Intelligence
Analytical Processing Pattern Recognition Predictive Modeling Automation	Empathy Self-Awareness Social Skills Relationship Management

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Hybrid Competencies

• Emotion-Aware AI Systems
• AI-Powered Feedback Tools
• Human-AI Conflict Resolution
• Culturally Adaptive AI
• Ethical AI Governance

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Enhanced Organizational Outcomes

Better Decision-Making
Improved Team Resilience
Human-Centric Workplaces

Figure 6: The Convergence of AI and Emotional Intelligence in Organizations (Text Representation)

and responsibility almost equally between management and workers (Taylor, 1911).

Anderson (Anderson, 1949) provides a thoughtful analysis of scientific management's meaning and implications. Writing thirty-eight years after Taylor's original publication, Anderson notes that scientific management is often misunderstood as merely a set of techniques for increasing efficiency. Instead, he argues that it represents a fundamental philosophy about the relationship between workers and management. The core insight is that both parties share a common interest in efficiency because higher productivity enables higher wages and profits simultaneously.

Critically, Anderson (Anderson, 1949) addresses the persistent criticism that scientific management dehumanizes workers by treating them as cogs in a machine. He acknowledges that Taylor's methods can be applied mechanistically but argues that this represents a misapplication of Taylor's philosophy. Properly understood, scientific management requires managers to study workers' ca-

Integrated Framework: Classical Management Meets Agentic AI

Classical Foundations	**Contemporary AI Scholarship**
Taylor's Scientific Management (Task Optimization) Fayol's Administrative Theory (Enterprise Coordination)	Joshi (2025a): AI in Management & Leadership Joshi (2025b): AI & Emotional Intelligence Joshi (2025c): AI in Teamwork & Resilience Joshi (2025d): Quantitative Models

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Taylor-Fayol Agent Framework

Taylor Agents (Task Optimizers)
Fayol Agents (Coordinators)

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Banking Applications
Algorithmic Trading — Risk Management
Customer Service — Compliance Monitoring

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Implications
Human Manager Evolution — Organizational Design
Risks & Limitations — Future Directions

Figure 7: Integrated Framework: Classical Management Meets Agentic AI in Quantitative Finance (Text Representation)

pabilities and design work that fits those capabilities, rather than forcing workers to adapt to poorly designed work processes.

3.2 Fayol's Administrative Theory

While Taylor analyzed work at the task level, Henri Fayol developed a comprehensive theory of management that addressed the entire organization. Fayol's experience as a mining executive gave him a different perspective—he observed what managers actually do and derived principles from those observations. According to Fayol, all managers perform five functions: planning, organizing, commanding, coordinating, and controlling (later condensed to planning, organizing, leading, and controlling in many modern textbooks).

Fayol also proposed fourteen principles of management that remain influential: division of work, authority and responsibility, discipline, unity of command, unity of direction, subordination of individual interests to general interests, remuneration, centralization, scalar chain, order, equity, stability of tenure, initiative, and esprit de corps (Fayol, 1949). These principles provide practical guidance for managers facing the complex challenge of coordinating human effort toward organizational goals.

The Medcrine (Medcrine, 2022) video on Fayol's fourteen principles illustrates how these concepts apply in contemporary organizations. Division of work, for example, remains fundamental to organizational design, though specialization now extends far beyond the manufacturing context Fayol envisioned. Similarly, unity of command—the principle that each employee should receive orders from only one superior—continues to influence organizational structure despite challenges from matrix organizations and team-based structures.

3.3 Complementarity and Integration

Taylor and Fayol's works complement each other in several important ways. First, they address different levels of analysis. Taylor's scientific management operates at the task level, providing methods for optimizing how individual jobs are performed. Fayol's administrative theory operates at the organizational level, providing frameworks for coordinating multiple tasks and functions toward common goals. A complete theory of management requires both perspectives—optimizing tasks without coordinating them leads to suboptimization, while coordinating without optimizing leads to inefficiency.

Second, Taylor and Fayol emphasize different mechanisms of control. Taylor relies on measurement and standardization—by scientifically determining the one best way to perform a task and training workers to follow that method, managers can ensure efficiency. Fayol relies on structural mechanisms—clear lines of authority, defined responsibilities, and coordinated planning—to align individual efforts with organizational objectives. Modern organizations combine both approaches, using measurement to optimize processes while maintaining structural clarity about who is responsible for what.

Third, Taylor and Fayol have different implications for the manager's role. In Taylor's framework, managers are scientists who study work, design optimal methods, and train workers to follow those methods. In Fayol's framework, managers are coordinators who plan, organize, and control the activities of others. McGregor's Theory X and Theory Y, mentioned in the module introduction, capture this tension: Taylor's assumptions about workers requiring close supervision align with Theory X, while Fayol's emphasis on initiative and esprit de corps aligns more closely with Theory Y.

4 Literature Review

The required readings for this module extend classical management theory in several important directions. Mintzberg (Mintzberg, 1990) challenges the folklore about what managers do, arguing that the classical functions described by Fayol do not adequately capture the reality of managerial work. Through observational research, Mintzberg found that managers work at an unrelenting pace, engage in activities characterized by brevity, variety, and fragmentation, and prefer verbal media to written communication. He identified ten managerial roles grouped into three categories: interpersonal roles (figurehead, leader, liaison), informational roles (monitor, disseminator, spokesperson), and decisional roles (entrepreneur, disturbance handler, resource allocator, negotiator).

Mintzberg's (Mintzberg, 1990) analysis does not reject Fayol's framework but rather supplements it. The classical functions describe what managers aim to accomplish, while Mintzberg's roles describe how managers actually spend their time. Both perspectives are necessary for understanding management. When managers plan, they do so through the activities Mintzberg describes—gathering information, allocating resources, negotiating with stakeholders. When they organize, they perform interpersonal roles that build the relationships necessary for coordinated action.

Gherson and Gratton (Gherson & Gratton, 2022) address a contemporary challenge: managers are expected to do too much. The authors argue that organizations have added responsibilities to managers' roles—coaching employees, fostering inclusion, supporting wellbeing, driving innovation—without removing anything. This role expansion leads to burnout and reduces effectiveness. Their solution involves redistributing managerial responsibilities across teams, leveraging peer support, and using technology to automate routine tasks. This analysis connects directly to both Taylor and Fayol. Taylor would recognize the need to analyze managerial work scientifically to identify what tasks require managerial attention versus what can be delegated or automated. Fayol would recognize the organizational design challenge of redefining roles and responsibilities.

Krishnamoorthy (Krishnamoorthy, 2022) examines what great remote managers do differently, a topic that would have seemed foreign to Taylor and Fayol but is essential in contemporary organizations. Remote managers excel by focusing on outcomes rather than activities, communicating with clarity and frequency, building trust through consistency, and adapting their style to individual team members' needs. These findings align with Fayol's principles of unity of direction and equity while challenging Taylor's emphasis on close supervision and standardized methods.

Ocampo, Gu, and Heyden (Ocampo, Gu, & Heyden, 2022) explore the costs of perfectionist management, finding that managers' high expectations can actually improve team performance when managed appropriately. The key is distinguishing between excellence—pursuing high standards through learning and growth—and perfectionism—demanding flawlessness and reacting negatively to mistakes. This distinction has implications for how Taylor's scientific management is applied. A perfectionist application seeks to eliminate all variation and enforce rigid compliance. An excellence-oriented application uses measurement to enable learning and continuous improvement.

Chang et al. (Chang et al., 2023) address how managers can incorporate diversity, equity, and inclusion (DEI) into decision-making. Their research demonstrates that DEI improves when managers are prompted to consider it at critical moments such as hiring, promotions, and performance reviews. This finding connects to both Taylor and Fayol. Taylor's emphasis on systematic, data-driven decision-making can be applied to DEI by measuring outcomes and identifying bias.

Fayol's principles of equity and justice directly address fair treatment of employees.

Field, Hancock, and Schaninger (Field, Hancock, & Schaninger, 2023) argue against the trend of eliminating middle managers, presenting evidence that middle managers create significant value when properly supported. They find that middle managers translate strategy into execution, coach and develop employees, and coordinate across organizational boundaries. This argument echoes Fayol's emphasis on the scalar chain and unity of command while recognizing that managers at all levels perform essential coordination functions.

Hocking (Hocking, 2024) addresses a practical challenge for first-time managers: learning to delegate. The article provides guidance on overcoming the tendency to hold onto tasks rather than trusting others to perform them. This connects directly to Taylor's principle of division of labor—managers must focus on their unique contributions while delegating tasks that others can perform. It also connects to Fayol's principle of authority and responsibility—delegation requires giving others both the authority to act and the responsibility for outcomes.

The required videos provide accessible introductions to key concepts. Hinshelwood (Hinshelwood, 2023) explains Taylorism and its influence on project management, tracing connections between scientific management and modern methodologies like critical path analysis and earned value management. Simplilearn (Simplilearn, 2022) distinguishes leadership from management, a distinction that will be explored further in Module 2. The Business Professor (Professor, 2022) provides a brief introduction to organizational behavior as a field of study.

Optional readings extend these themes. Pistrui and Dimov (Pistrui & Dimov, 2018) argue that the manager's role must change in five key ways to address contemporary challenges. Soren (Soren, 2017) provides a comprehensive review of Fayol's influence over 100 years, demonstrating the enduring relevance of his framework.

4.1 Contemporary AI Scholarship in Management

Recent scholarship has synthesized classical management themes with the AI revolution in organizations. Joshi (Joshi, 2025d) provides a comprehensive review of artificial intelligence in management, leadership, decision-making, and collaboration, offering an organizational behavior-focused synthesis that bridges classical management theory with contemporary AI applications. This work examines how AI systems are reshaping fundamental management processes and identifies the capabilities managers need to develop in AI-augmented organizations.

The role of AI in enhancing teamwork, resilience, and decision-making has been systematically analyzed (Joshi, 2025e), revealing several key findings. Hybrid AI-human decision models achieve 38% faster response times while maintaining 89% prediction accuracy in behavioral assessments. The research introduces methodologies for enhancing team resilience through adaptive AI systems, cross-training interventions, and pre-mortem simulation techniques. A novel cultural alignment metric (C_{diff}) for global teams is proposed, and four STAR framework variants for AI-enhanced workforce development are evaluated. The study demonstrates how AI cognitive scaffolding systems can improve team antifragility when combined with human emotional intelligence.

The convergence of artificial intelligence and emotional intelligence has profound implications for leadership and organizational behavior (Joshi, 2025a). This research identifies key opportunities where AI can augment human emotional intelligence capabilities through emotion recognition systems and AI-powered feedback tools. Simultaneously, the paper explores how emotionally intelligent leadership remains essential for guiding ethical AI implementation and maintaining human-centric workplaces. The research highlights the growing importance of hybrid competencies that combine technical AI fluency with advanced emotional intelligence skills, particularly in areas like conflict resolution, team motivation, and change management. Challenges addressed include privacy concerns in emotion-aware technologies, the risk of over-reliance on automated systems, and the need for cultural adaptation in global organizations.

Joshi (Joshi, 2025b) systematically reviews current research on AI's role in leadership, identifying three key areas of impact: (1) enhanced strategic decision-making through human-AI collaboration, (2) evolution of leadership styles in digital environments, and (3) organizational challenges in AI adoption. The study reveals significant improvements in decision accuracy and speed when combining AI tools with human judgment, while also highlighting critical gaps in implementation including cultural adaptation, ethical governance, and long-term effectiveness measurement. Practical frameworks for AI integration in leadership are proposed, emphasizing the balance between technological capabilities and human-centered values.

Quantitative models and visualization approaches for managerial decision-making in the age of AI are reviewed in (Joshi, 2025c), demonstrating how data-driven methods can support and enhance managerial judgment. This work provides essential context for understanding how Taylor's emphasis on measurement and Fayol's focus on coordination can be operationalized in AI-driven environments.

These analyses together provide essential context for understanding how Taylor and Fayol's frameworks apply in AI-driven environments and inform the framework developed in the next section.

5 Agentic Generative AI in Banking: A Taylor-Fayol Framework

The emergence of agentic generative AI represents a paradigm shift in how banking and financial services operate. Unlike traditional AI systems that execute predefined rules or simple automations, agentic AI systems possess autonomy, goal-directed behavior, and the ability to interact with other agents and humans. This section introduces a novel framework for understanding these systems through the lens of Taylor and Fayol's complementary management theories, drawing on contemporary AI scholarship (Joshi, 2025d, 2025c, 2025e, 2025a, 2025b).

5.1 Conceptual Framework: Taylor Agents and Fayol Agents

In a multi-agent AI system, different agents can be designed to perform different management functions. Some agents specialize in optimizing specific tasks—these function as digital embodiments of Taylor's scientific management. Other agents specialize in coordinating across tasks, resolving conflicts, and ensuring alignment with organizational goals—these function as digital embodiments of Fayol's administrative theory. Table 1 summarizes this framework.

Table 1: Taylor Agents and Fayol Agents in Banking AI Systems

Agent Type	**Taylor Agents (Task Optimizers)**	**Fayol Agents (Coordinators)**
Primary Function	Optimize specific tasks through measurement and analysis	Coordinate across agents and align with organizational goals
Theoretical Basis	Scientific management (Taylor, 1911)	Administrative theory (Fayol, 1949)
Control Mechanism	Measurement, standardization, feedback loops	Communication protocols, hierarchy, shared objectives
Banking Applications	Algorithmic trading, credit scoring, fraud detection	Portfolio coordination, risk aggregation, compliance monitoring
Communication Pattern	Report results to Fayol agents; receive optimization targets	Receive reports; issue coordination directives; resolve conflicts
Autonomy Level	High within defined task boundaries	High across multiple functions

This framework recognizes that effective multi-agent systems, like effective organizations, require both specialized optimization and coordinative integration. Taylor agents ensure that individual tasks are performed efficiently; Fayol agents ensure that the overall system achieves its objectives without suboptimization or conflict. Recent research on AI in management has extensively documented how such distributed intelligence architectures are transforming organizational decision-making and collaboration (Joshi, 2025d, 2025c, 2025e, 2025a, 2025b). The integration of emotional intelligence with AI systems (Joshi, 2025a) further enhances the capabilities of both agent types, enabling more nuanced human-AI interaction.

5.2 Taylor Agents in Banking Operations

Taylor agents embody the principles of scientific management in specific banking functions. They measure, analyze, and optimize particular tasks with minimal human intervention. Several examples illustrate this application.

Algorithmic Trading Agents. In quantitative finance, trading agents function as pure Taylor agents. They continuously analyze market data, identify patterns, and execute trades according to mathematically optimized strategies. These agents embody Taylor's principle of replacing rule-of-thumb methods with scientific analysis. They measure everything—execution costs, market impact, timing precision—and adjust their behavior based on empirical feedback. Renaissance Technologies' Medallion Fund, perhaps the most successful quantitative fund in history, operates through such agents, executing millions of trades based on statistical patterns too subtle for humans to detect (Zuckerman, 2019). The quantitative models underlying these agents align with the decision-making frameworks reviewed in (Joshi, 2025c).

Credit Underwriting Agents. Banks increasingly use AI agents to evaluate loan applications. These agents analyze applicant data, assess creditworthiness, and make approval decisions based on predictive models. They embody Taylor's principle of scientifically selecting and training workers—in this case, the "worker" is the model itself, which is trained on historical data to optimize prediction accuracy. These agents continuously learn from new data, updating their models as economic conditions change and as they observe loan performance. The integration of emotional intelligence considerations (Joshi, 2025a) can help ensure that algorithmic decisions remain fair and equitable.

Fraud Detection Agents. Fraud detection systems operate as Taylor agents that monitor transactions in real-time, flagging suspicious activity based on statistical anomalies. They embody Taylor's principle of measurement and standardization—normal transaction patterns are established empirically, and deviations trigger investigation. These agents operate at scale impossible for human analysts, reviewing millions of transactions per second and identifying fraud patterns that would escape human notice.

Customer Service Agents. Generative AI chatbots now handle a significant portion of customer service interactions in banking. These agents function as Taylor agents for specific service tasks—answering account balance inquiries, processing simple transactions, explaining products.

They optimize for resolution speed and customer satisfaction, measuring their performance and learning from each interaction. The integration of emotional intelligence capabilities (Joshi, 2025a) enables these agents to better recognize and respond to customer emotions.

5.3 Fayol Agents: Coordination in Multi-Agent Systems

As organizations deploy multiple Taylor agents, the need for coordination becomes critical. Different agents may pursue conflicting objectives, compete for resources, or generate outputs that require integration. Fayol agents address these coordinative challenges, embodying the leadership functions identified in contemporary AI scholarship (Joshi, 2025b, 2025d).

Portfolio Coordination Agents. In a quantitative fund, multiple trading agents may operate simultaneously, each pursuing different strategies in different markets. A portfolio coordination agent functions as a Fayol agent, ensuring that the collective positions remain within risk limits, that correlated strategies do not create unintended concentrations, and that capital is allocated optimally across strategies. This agent embodies Fayol's planning and coordinating functions—it sets boundaries within which Taylor agents operate and adjusts those boundaries based on overall portfolio performance.

Risk Aggregation Agents. Banks maintain multiple risk management systems monitoring different risk types—market risk, credit risk, operational risk, liquidity risk. A risk aggregation agent functions as a Fayol agent that consolidates these diverse risk measures into an integrated view. It applies Fayol's principle of unity of direction, ensuring that all risk management activities align with the bank's overall risk appetite. When the aggregated risk exceeds thresholds, this agent may issue directives to Taylor agents to reduce positions or adjust exposures.

Compliance Monitoring Agents. Regulatory compliance requires coordinating multiple systems and processes to ensure that all activities conform to legal requirements. Compliance agents function as Fayol agents that monitor the outputs of Taylor agents, flag potential violations, and ensure that corrective actions are taken. They embody Fayol's controlling function, verifying that actual performance conforms to plans and standards. Ethical governance frameworks (Joshi, 2025b) are essential for ensuring these agents operate appropriately.

Orchestration Agents. Complex banking processes often require multiple specialized agents to collaborate. For example, onboarding a new corporate client may involve agents for account setup, credit evaluation, compliance screening, and system configuration. An orchestration agent functions as a Fayol agent that sequences these activities, manages handoffs between agents, and ensures that the overall process completes successfully. This agent embodies Fayol's organizing function, structuring the workflow and defining how different agents interact.

5.4 Agent Communication and Fayol's Scalar Chain

One of Fayol's most enduring principles is the scalar chain—the line of authority from top to bottom of an organization. In multi-agent AI systems, this principle manifests in communication protocols and hierarchical relationships among agents.

Taylor agents typically communicate upward to Fayol agents, reporting their activities, performance metrics, and any exceptions requiring attention. Fayol agents communicate downward to Taylor agents, issuing directives, setting targets, and providing feedback. This hierarchical structure maintains order and prevents the confusion that would arise if all agents communicated with all others indiscriminately.

However, agentic systems also enable more flexible communication patterns than traditional organizations. Taylor agents may communicate laterally with each other when coordination is required—for example, a trading agent in equities may need to coordinate with a trading agent in derivatives to execute a complex strategy. This lateral communication, when authorized by Fayol agents, creates what Fayol called "gangplanks"—direct communication between peers that speeds decision-making while maintaining hierarchical integrity.

The Simplilearn (Simplilearn, 2022) video on leadership versus management becomes relevant here. In multi-agent systems, Taylor agents function more as managers—they optimize specific processes and ensure tasks are completed efficiently. Fayol agents function more as leaders—they set direction, align activities with purpose, and ensure that the whole is greater than the sum of the parts. Both are necessary, just as both management and leadership are necessary in human organizations. Joshi's reviews of leadership models in AI contexts (Joshi, 2025c, 2025b, 2025d) provide additional depth on how these roles evolve when intelligent systems share managerial responsibilities, while the integration of emotional intelligence (Joshi, 2025a) enhances the quality of human-agent interactions.

5.5 Case Illustration: AI-Driven Trade Execution

A concrete example illustrates how Taylor and Fayol agents collaborate in a quantitative trading context. Consider a hedge fund executing a complex trade—purchasing a large block of shares without moving the market against itself.

Taylor Agents:

- A **liquidity detection agent** analyzes order books to identify where shares can be purchased with minimal market impact
- An **execution timing agent** determines optimal times to enter the market based on historical patterns of volume and volatility
- A **venue selection agent** decides which exchanges or dark pools to use for each portion of the trade
- A **transaction cost analysis agent** measures actual execution costs and compares them to benchmarks

Fayol Agents:

- A **trade coordination agent** receives the overall trading objective, divides it among the Taylor agents, and monitors their progress
- A **risk management agent** ensures that positions taken during execution do not exceed intraday risk limits

- A **market impact agent** aggregates the activities of all Taylor agents to estimate total footprint and adjusts the execution schedule if impact threatens to become excessive

These agents communicate continuously throughout the execution process. Taylor agents report their planned actions and seek confirmation. The trade coordination agent ensures that their activities are sequenced appropriately and that they do not compete against each other. The risk management agent monitors aggregate exposure and may pause execution if limits are approached. The market impact agent synthesizes information from all sources and may recommend slowing down if the trade is moving the market. The decision-making processes involved align with the hybrid AI-human models analyzed in recent scholarship (Joshi, 2025e, 2025c).

This multi-agent architecture achieves what neither Taylor nor Fayol could have imagined: autonomous, coordinated action toward complex goals, with specialized agents optimizing individual tasks while coordinative agents ensuring overall effectiveness. Yet the underlying principles—specialization and coordination, measurement and alignment—derive directly from their complementary frameworks.

6 Implications for Management in Quantitative Finance

The rise of agentic generative AI in banking has profound implications for how organizations are managed and how managers perform their roles. This section examines these implications through the lens of the Taylor-Fayol framework, drawing on contemporary AI scholarship (Joshi, 2025d, 2025c, 2025e, 2025a, 2025b).

6.1 The Changing Role of Human Managers

As AI agents take on both specialized and coordinative functions, the role of human managers transforms. Managers are no longer primarily responsible for directing and controlling workers; instead, they are responsible for designing, deploying, and overseeing agentic systems. This shift requires new capabilities and new ways of thinking about management, aligning with the evolving leadership paradigms identified in recent research (Joshi, 2025b).

System Designers. Human managers must become system designers who architect the relationships between Taylor agents and Fayol agents. They must determine what tasks should be automated, what coordination mechanisms should be established, and how agents should communicate with each other and with humans. This design work requires understanding both the capabilities of AI systems and the requirements of the business. Joshi's synthesis of AI in management (Joshi, 2025d) provides a comprehensive overview of how managers can develop these design capabilities, while research on AI-enhanced teamwork (Joshi, 2025e) offers insights into effective human-AI collaboration structures.

Boundary Setters. Managers must set boundaries within which agents operate autonomously. These boundaries include risk limits, compliance constraints, and strategic guidelines. Setting appropriate boundaries requires judgment about what agents can handle independently versus what requires human oversight. It also requires anticipating edge cases where agents might behave unexpectedly. The ethical governance frameworks discussed in (Joshi, 2025b) are essential for this boundary-setting function.

Exception Handlers. Despite sophisticated AI, exceptions will occur—market conditions that models have not seen, conflicts that agents cannot resolve, situations requiring human judgment. Managers must be prepared to step in, understand what has happened, and take appropriate action. This exception-handling role requires deep domain knowledge and the ability to make decisions under pressure, drawing on the hybrid decision-making models analyzed in (Joshi, 2025e, 2025c).

Performance Evaluators. Managers must evaluate how well agentic systems are performing, both individually and collectively. Are Taylor agents achieving their optimization objectives? Are Fayol agents coordinating effectively? Is the overall system delivering the intended results? These evaluations require metrics that capture both efficiency and effectiveness, much as Taylor advocated measuring worker performance. Quantitative models and visualization tools can support these evaluations, as reviewed in recent scholarship (Joshi, 2025c).

Emotional Intelligence Leaders. As AI systems take on more analytical tasks, the uniquely human capabilities of emotional intelligence become increasingly valuable (Joshi, 2025a). Managers must model and foster empathy, self-compassion, and emotional awareness within their teams, creating environments where humans and AI systems can collaborate effectively. The convergence of AI and emotional intelligence (Joshi, 2025a) suggests that the most effective future leaders will be those who can harness both technological and human capabilities.

Krishnamoorthy's (Krishnamoorthy, 2022) insights about remote management apply here: managers must focus on outcomes rather than activities, communicate with clarity and frequency, and build trust through consistency. When managing AI agents, trust comes from understanding how agents work, validating their outputs, and observing their performance over time.

6.2 Organizational Design Implications

The Taylor-Fayol agent framework has implications for how financial organizations structure themselves. Traditional organizational charts depict reporting relationships among humans. Future organizations may need charts that depict relationships among agents and between agents and humans, reflecting the distributed intelligence architectures documented in recent research (Joshi, 2025d).

Hybrid Hierarchies. Organizations will develop hybrid hierarchies where humans supervise Fayol agents, Fayol agents supervise Taylor agents, and Taylor agents execute tasks. This hierarchy maintains Fayol's scalar chain while leveraging AI at each level. The challenge is ensuring that authority and responsibility are clearly defined at each level and that humans maintain ultimate control. The leadership frameworks discussed in (Joshi, 2025b) provide guidance for designing these hierarchies.

Agent Teams. Just as human organizations use cross-functional teams, agentic systems may deploy teams of specialized agents that collaborate on complex tasks. These teams may include both Taylor agents (task specialists) and Fayol agents (coordinators) that work together without constant human oversight. Managing these agent teams requires new approaches to team design and performance management. Research on AI-enhanced collaboration (Joshi, 2025d, 2025e) provides guidance on these emerging organizational forms, while insights from emotional intelligence research (Joshi, 2025a) can inform the design of more effective human-agent teams.

Human-Agent Collaboration. Many tasks will be performed through collaboration between humans and agents rather than by either alone. For example, a portfolio manager may work with a suite of agents that analyze opportunities, assess risks, and propose trades. The manager makes final decisions but relies on agents for analysis and execution. This collaboration requires interfaces that enable effective communication and shared understanding between humans and agents. The hybrid decision-making models analyzed in (Joshi, 2025e, 2025c) demonstrate that such collaboration can achieve superior outcomes when properly designed.

Gherson and Gratton's (Gherson & Gratton, 2022) argument that managers can't do it all applies here. By delegating both specialized tasks and coordinative functions to AI agents, organizations can free human managers to focus on higher-value activities—strategy, innovation, relationship-building, and judgment in ambiguous situations. The key is thoughtful redistribution of responsibilities, not simply adding AI on top of existing managerial workloads.

6.3 Risks and Limitations

The agentic AI approach also introduces risks that managers must address. These risks echo concerns that critics have raised about Taylorism while also introducing new challenges specific to AI. Recent scholarship has extensively documented these challenges (Joshi, 2025b, 2025d, 2025a).

Suboptimization. Taylor agents optimized for specific tasks may pursue their objectives in ways that harm overall system performance. A trading agent optimized for short-term profits may take excessive risk that concerns the portfolio as a whole. A customer service agent optimized for speed may provide incorrect information that creates downstream problems. Fayol agents mitigate this risk through coordination, but the risk cannot be eliminated entirely.

Black Box Problems. As agents become more sophisticated, understanding why they made particular decisions becomes more difficult. A Taylor agent that denies a loan application may have considered hundreds of variables in ways humans cannot easily reconstruct. This black box problem complicates oversight and makes it difficult to ensure fairness and compliance. The ethical governance frameworks discussed in (Joshi, 2025b) are essential for addressing these challenges.

Loss of Human Judgment. Over-reliance on AI agents may erode human judgment and expertise. If portfolio managers always defer to agent recommendations, they may lose the ability to make independent judgments when agents fail or when unprecedented situations arise. Maintaining human judgment alongside AI capabilities is essential. Joshi's analysis of decision-making models (Joshi, 2025c) emphasizes the importance of keeping humans in the loop for critical judgments, while research on emotional intelligence (Joshi, 2025a) highlights the uniquely human capabilities that remain essential.

Privacy and Ethical Concerns. Emotion-aware AI systems and other advanced applications raise significant privacy concerns (Joshi, 2025a). Organizations must establish clear policies regarding data collection, usage, and storage to maintain trust and comply with regulations.

Ocampo, Gu, and Heyden's (Ocampo et al., 2022) distinction between excellence and perfectionism is relevant here. Pursuing perfect AI systems that never make mistakes may lead to over-engineering, brittleness, and inability to handle novel situations. Pursuing excellence—continuous improvement, learning from mistakes, maintaining human oversight—is more likely to produce sustainable results.

6.4 Future Directions

The agentic AI revolution in banking is just beginning. Several trends will shape its evolution in coming years, as identified in recent scholarship (Joshi, 2025b, 2025d, 2025a, 2025e).

More Sophisticated Fayol Agents. As multi-agent systems become more common, Fayol agents will become more sophisticated. They may develop capabilities for strategic planning, resource allocation across multiple objectives, and even designing and deploying new Taylor agents as needs emerge. These agents will increasingly perform functions that today require human managers.

Enhanced Emotional Intelligence Capabilities. Future AI systems will incorporate increasingly sophisticated emotional intelligence capabilities (Joshi, 2025a), enabling more natural and effective human-AI interaction. Emotion recognition systems, AI-powered feedback tools, and affective computing will become standard components of agentic systems.

Human-Agent Teams. Rather than agents replacing humans or humans supervising agents, we will see more true teamwork where humans and agents collaborate as peers. This collaboration will require agents that can explain their reasoning, accept feedback, and adapt to human preferences. It will require humans who understand how to work effectively with intelligent systems. Recent scholarship on AI in collaboration (Joshi, 2025d, 2025e) provides frameworks for understanding these emerging team dynamics.

Agentic Governance. Organizations will need governance frameworks for their agentic systems. Who is responsible when agents make mistakes? How are agent objectives aligned with organizational values? How are conflicts between agents resolved? These governance questions will become as important as traditional corporate governance. The ethical frameworks discussed in (Joshi, 2025b) provide starting points for developing such governance structures.

Field, Hancock, and Schaninger's (Field et al., 2023) defense of middle managers suggests that even as AI agents take on more functions, human managers will remain essential—not for routine direction and control, but for the uniquely human capabilities of judgment, wisdom, responsibility, and emotional intelligence (Joshi, 2025a). The challenge is redefining their roles for a world where many traditional management functions are performed by intelligent systems.

7 Conclusion

The classical management frameworks of Frederick Winslow Taylor and Henri Fayol remain essential foundations for understanding management in contemporary American financial institutions, including those transformed by agentic generative AI. As US banks, hedge funds, and financial services firms compete in an increasingly global and technology-driven marketplace, the principles established by these foundational thinkers offer critical guidance for maintaining competitive advantage. Taylor's scientific management provides tools for optimizing tasks through systematic measurement and analysis—principles that now manifest in specialized AI agents optimizing trading, lending, fraud detection, and customer service across American financial institutions. Fayol's administrative theory provides frameworks for coordinating multiple activities toward common goals—principles that now manifest in coordinative AI agents that integrate specialized agents, manage risks, ensure compliance with US regulations, and orchestrate complex financial processes.

The metaphor of Taylor agents and Fayol agents captures this complementarity. Taylor agents optimize individual tasks through measurement and feedback, embodying the scientific management tradition that helped build America's industrial dominance. Fayol agents coordinate across agents and align activities with organizational objectives, embodying the administrative theory tradition essential for managing complex enterprises. Together, they create multi-agent systems that achieve both efficiency and integration—the twin requirements that Taylor and Fayol identified for effective organizations and that remain critical for American financial competitiveness.

Recent scholarship has extensively documented how these dynamics play out in AI-augmented organizations. Research has examined AI's role in enhancing teamwork, resilience, and decision-making (Joshi, 2025e), demonstrating that hybrid AI-human decision models achieve significant improvements in response times while maintaining high prediction accuracy. Studies on AI in leadership and management (Joshi, 2025b) have identified three key areas of impact: enhanced strategic decision-making through human-AI collaboration, evolution of leadership styles in digital environments, and organizational challenges in AI adoption. The convergence of artificial intelligence and emotional intelligence (Joshi, 2025a) has profound implications for leadership and organizational behavior, highlighting the growing importance of hybrid competencies that combine technical AI fluency with advanced emotional intelligence skills—capabilities essential for American managers leading AI-transformed organizations. Quantitative models and visualization approaches for managerial decision-making (Joshi, 2025c) provide tools for implementing these frameworks in practice. Comprehensive reviews of AI in management, leadership, decision-making, and collaboration (Joshi, 2025d) offer organizational behavior-focused syntheses that bridge classical management theory with contemporary AI applications.

In American quantitative finance, these agentic systems are transforming how financial institutions operate. Trading at major US investment banks is increasingly performed by autonomous agents that analyze markets and execute strategies. Risk management across American financial holding companies is performed by agents that monitor exposures and ensure compliance with Federal Reserve and SEC requirements. Customer service at US retail banks is performed by agents that handle routine inquiries and escalate complex issues. And coordination among these specialized agents is performed by higher-level agents that embody Fayol's management functions—creating a new organizational paradigm for American finance.

For American managers, this transformation requires new capabilities and new ways of thinking. Managers must become system designers, boundary setters, exception handlers, performance evaluators, and emotional intelligence leaders. They must maintain human judgment while leveraging machine intelligence. They must ensure that agentic systems serve organizational purposes and comply with US financial regulations rather than operating in isolation or pursuing narrow objectives at the expense of the whole. The integration of emotional intelligence (Joshi, 2025a) is essential for maintaining human-centric workplaces in an increasingly automated world—a critical consideration for American firms competing for talent.

Several limitations of this analysis suggest directions for future research with particular relevance to US financial institutions. First, the framework proposed here requires empirical validation through research examining how actual multi-agent systems at American banks align with the Taylor-Fayol distinction. Second, the ethical implications of autonomous agents making financial decisions deserve deeper examination, particularly regarding compliance with US consumer protection laws, fairness in lending, and accountability under American regulatory frameworks—building on the governance frameworks discussed in (Joshi, 2025b). Third, the evolution of human-agent collaboration over time—how trust develops, how roles shift, how conflicts are resolved—merits longitudinal study at American financial institutions, drawing on insights from teamwork research (Joshi, 2025e) and emotional intelligence scholarship (Joshi, 2025a). Fourth, the quantitative models and visualization tools for managing AI-augmented organizations require further development and testing (Joshi, 2025c) in the context of US financial markets. Fifth, the integration of emotional intelligence capabilities into agentic systems and the development of hybrid human-AI emotional competencies (Joshi, 2025a) represent important frontiers for future research that could strengthen American leadership in financial technology.

The module introduction noted that Taylor's work was voted the most influential management book of the twentieth century in 2001. As America progresses through the twenty-first century, Taylor and Fayol's influence persists, but their frameworks require reconceptualization for a world where intelligent agents, rather than humans, perform many management functions within US financial institutions. The Taylor-Fayol agent framework proposed here represents one such reconceptualization—an attempt to map classical principles onto emerging organizational forms that will determine America's competitive position in global finance. The challenge for contemporary American managers is to understand these foundations while building the new capabilities that agentic AI demands, including the hybrid competencies that combine technical AI fluency with advanced emotional intelligence (Joshi, 2025a). American financial institutions that successfully integrate AI's analytical power with human emotional wisdom will be best positioned to thrive in the age of quantitative finance and agentic AI, securing the nation's leadership in global financial markets for decades to come.

Compliance & Ethical Considerations

The views expressed in this paper are solely those of the author and do not necessarily represent the views of any affiliated institutions. This work was conducted as part of independent research and academic coursework in partial fulfillment of DBA701: Theories of Organizational Behavior in Business.

This is a review paper that synthesizes and organizes existing research from the cited literature. All findings, proposals, and frameworks presented herein are derived from previously published sources as referenced throughout the document. The author's original contribution lies in the synthesis, organization, and conceptual integration of these existing materials, particularly the novel Taylor-Fayol agent framework proposed in Section 4.

Portions of this manuscript were drafted with the assistance of AI writing tools (including ChatGPT and Claude by Anthropic) to improve clarity, organization, and flow. All AI-generated content was carefully reviewed, edited, and verified by the author for coherence, factual accuracy, and alignment with cited sources. Reasonable efforts were made to identify and eliminate potential hallucinations or inaccuracies. The LaTeX code was developed with the assistance of GitHub Copilot and edited through DeepSeek.

Final responsibility for the content, including any errors, omissions, or interpretations, rests with the author. Readers are encouraged to validate all claims against the original cited sources and draw their own conclusions..

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